

SCIENCE EDUCATION

PHYSICS

A Comprehensive Guide for Class XI

Premium Edition • CBSE Curriculum

Motion in a Straight Line

Understanding Position, Velocity, and Acceleration in One Dimension

Motion is one of the most fundamental phenomena we observe in our daily lives. From the moment you wake up and reach for your alarm clock, to commuting to school, playing sports, or watching celestial bodies move across the sky — motion is everywhere. Understanding motion scientifically allows us to predict where objects will be at any given time, how fast they're moving, and how their speed changes.

Kinematics is the branch of mechanics that describes motion without considering what causes it. In this chapter, we focus on the simplest type of motion: motion in a straight line, also called **one-dimensional** or **rectilinear motion**. This forms the foundation for understanding more complex motions in two and three dimensions.

The beauty of studying motion in one dimension is that we can use simple positive and negative signs to indicate direction, making our mathematical treatment straightforward. By the end of this chapter, you'll be able to analyze the motion of cars on highways, falling objects, and any other object moving along a straight path.

Historical Background: Understanding Motion Through the Ages

Aristotle (384–322 BC) dominated thinking about motion for nearly 2000 years. He divided motion into "natural" (objects seeking their natural place — heavy things fall, smoke rises) and "violent" (forced motion, like throwing a stone). He believed heavier objects fall faster and that a force is needed to *maintain* motion. These ideas, based on philosophical reasoning rather than experiments, went essentially unchallenged until the Renaissance.

Galileo Galilei (1564–1642) is rightly called the *Father of Modern Kinematics*. Through brilliant experiments with inclined planes (around 1604), he became the first to systematically use mathematics to describe motion. He discovered that displacement is proportional to the square of time for uniformly accelerated motion ($s \propto t^2$), and that all objects fall with the same acceleration regardless of mass. Most importantly, he realized that no force is needed to maintain uniform motion — only to *change* it — laying the groundwork for Newton's First Law.

Isaac Newton (1642–1727) built on Galileo's insights and formalized kinematics within his monumental *Principia Mathematica* (1687). He developed **calculus** specifically to describe continuously changing motion — concepts like instantaneous velocity (dx/dt) and acceleration (d^2x/dt^2) were impossible to define rigorously without it.

Gottfried Wilhelm Leibniz (1646–1716) independently developed calculus around the same time as Newton. The dx/dt notation we use today for instantaneous velocity comes from Leibniz, while Newton used a dot notation ($\dot{x}$). Their bitter priority dispute lasted decades, but both deserve equal credit for giving us the mathematical language of motion.

2.1 Motion Basics

2.1.1 Rest and Motion

Definitions

Rest: An object is said to be at rest if it does not change its position with respect to its surroundings over time.

Motion: An object is said to be in motion if it changes its position with respect to its surroundings over time.

Rest and motion are **relative** concepts — they depend on the observer. A passenger sitting in a moving train is at rest relative to other passengers, but in motion relative to a person standing on the platform. There is no absolute rest or absolute motion in the universe.

💡 Illustrative Example

Consider a person sitting in an airplane flying at 900 km/h:

- Relative to the airplane → the person is at **rest**
- Relative to the ground → the person is in **motion** at 900 km/h
- Relative to the Sun → the person is in motion (Earth orbits at ~108,000 km/h!)

This shows that the same object can be at rest AND in motion simultaneously, depending on the observer.



Galileo Galilei — Father of Modern Kinematics

1564–1642, Italy

Galileo was the first to challenge Aristotle's 2000-year-old ideas about motion through careful experiments. Using inclined planes to "slow down" gravity and make measurements feasible, he precisely determined how distance varies with time and discovered that $s \propto t^2$ for uniformly accelerated motion. He also realized that in the absence of friction, a moving object would continue moving forever — the seed of Newton's First Law. His approach of combining experiment with mathematical analysis established the modern scientific method in physics.

2.1.2 Frame of Reference

📘 Definition

Frame of Reference: A coordinate system together with a clock, attached to an observer, used to describe the position and motion of objects. It consists of (i) an origin (reference point), (ii) a set of coordinate axes, and (iii) a time-measuring device (clock).

To describe the motion of an object, we must first choose a frame of reference. In one-dimensional motion, we typically choose an x -axis along the line of motion, with an origin at a convenient point. The position of the object is then described by a single coordinate x , which can be positive or negative depending on which side of the origin the object is located.

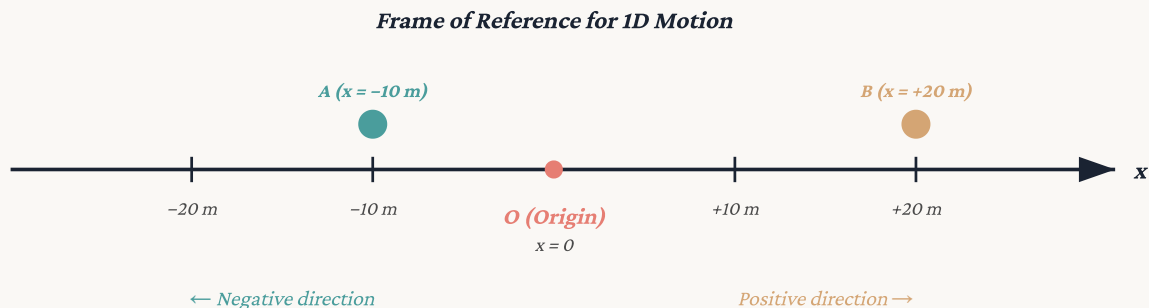


Figure 2.1: A one-dimensional frame of reference with origin O and objects at positions A and B

Types of Motion

- **Rectilinear motion:** Motion along a straight line (e.g., car on a highway, free fall)
- **Circular motion:** Motion along a circular path (e.g., hands of a clock, Earth around Sun)
- **Oscillatory motion:** To-and-fro motion about a fixed point (e.g., pendulum, vibrating string)
- **Random motion:** Irregular, unpredictable motion (e.g., Brownian motion, buzzing fly)

A **point object** (or particle) is an idealization where the size of the object is negligible compared to the distances involved in its motion. For example, a car traveling between two cities can be treated as a point object.

2.2 Position and Displacement

Core Definitions

Position (x): The location of an object at a given time, measured from a reference point (origin). It is described by a coordinate on the chosen axis and can be positive or negative.

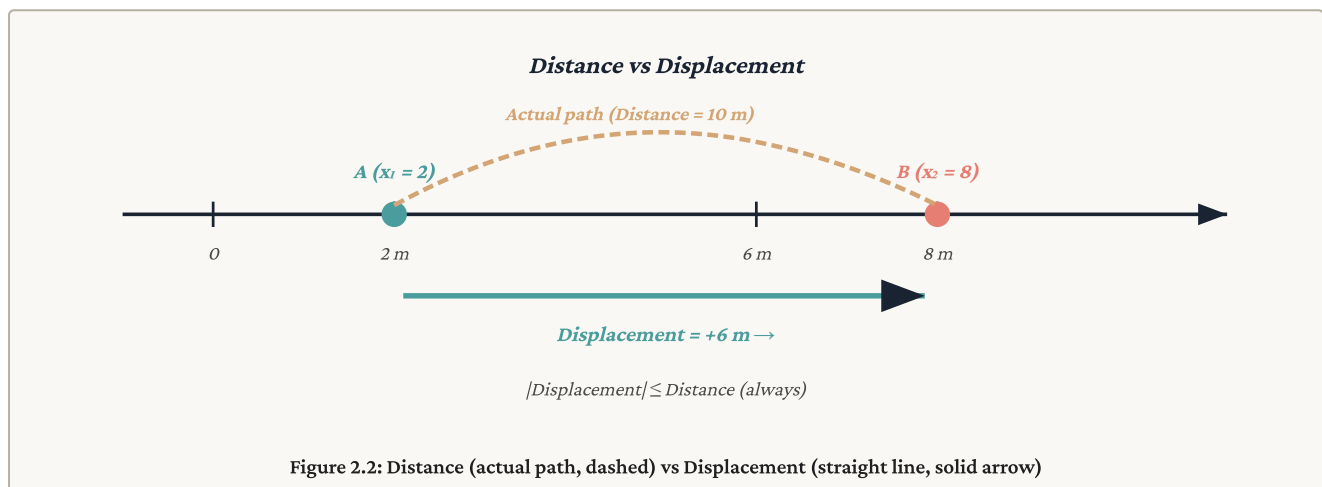
Distance: The total length of the path actually traveled by an object, regardless of direction. It is a *scalar* quantity (always ≥ 0).

Displacement (Δx): The change in position of an object. It is the shortest straight-line distance from the initial position to the final position, with direction. It is a *vector* quantity (can be +, -, or 0).

DISPLACEMENT FORMULA

$$\Delta x = x_2 - x_1 = x_{\text{final}} - x_{\text{initial}}$$

2.2.1 Distance vs Displacement



Solved Example 2.1

Q: A person walks 300 m east, then 400 m north. Find the distance traveled and displacement.

Solution:

Distance = 300 + 400 = **700 m**

Displacement = $\sqrt{(300^2 + 400^2)} = \sqrt{(90000 + 160000)} = \sqrt{250000} = \mathbf{500\ m}$ (at an angle to the east)

Here, distance > |displacement|, as expected.

2.2.2 Sign Convention in 1D Motion

In one-dimensional motion, direction is represented by algebraic signs. We adopt a convention and stick to it throughout a problem.

Standard Sign Convention

- **Horizontal motion:** Right = positive (+), Left = negative (-)
- **Vertical motion:** Upward = positive (+), Downward = negative (-)
- Displacement in the positive direction → $\Delta x > 0$
- Displacement in the negative direction → $\Delta x < 0$
- The choice of positive direction is arbitrary, but once chosen, must be used consistently

Property	Distance	Displacement
Type	Scalar	Vector
Sign	Always ≥ 0	Can be +, -, or 0

Property	Distance	Displacement
Depends on	Actual path taken	Only initial & final positions
For one-way motion	= Displacement	= $\pm$ Distance
For round trip	> 0	= 0
SI unit	metre (m)	metre (m)
Can decrease?	No (only increases or stays same)	Yes

2.3 Speed and Velocity

Just as we distinguished between distance and displacement, we must distinguish between speed and velocity. Both describe how fast an object is moving, but velocity includes directional information.

2.3.1 Average Speed and Average Velocity

Definitions

Average Speed: The total distance traveled divided by the total time taken. It is a scalar quantity and is always positive.

Average Velocity: The displacement divided by the time interval. It is a vector quantity and can be positive, negative, or zero.

SPEED AND VELOCITY FORMULAS

$$\text{Average Speed} = \text{Total Distance} / \text{Total Time} = d / t$$

$$\text{Average Velocity} = \text{Displacement} / \text{Time} = \Delta x / \Delta t = (x_2 - x_1) / (t_2 - t_1)$$

Special Cases for Average Speed

→ **Equal distances at different speeds:** If an object covers the first half of distance at speed v_1 and second half at speed v_2 , then:

$$\text{Average speed} = 2v_1v_2 / (v_1 + v_2) \text{ (harmonic mean)}$$

→ **Equal times at different speeds:** If an object moves at speed v_1 for time t and at speed v_2 for the same time t , then:

$$\text{Average speed} = (v_1 + v_2) / 2 \text{ (arithmetic mean)}$$

Solved Example 2.2

Q: A car travels from city A to city B (120 km) at 60 km/h and returns at 40 km/h. Find (a) average speed and (b) average velocity for the round trip.

Solution:

$$\text{Time for A} \rightarrow \text{B} = 120/60 = 2 \text{ h}$$

$$\text{Time for B} \rightarrow \text{A} = 120/40 = 3 \text{ h}$$

$$(a) \text{ Average speed} = \text{Total distance} / \text{Total time} = (120 + 120) / (2 + 3) = 240/5 = \mathbf{48 \text{ km/h}}$$

$$\text{Alternatively, using harmonic mean (equal distances): } 2 \times 60 \times 40 / (60 + 40) = 4800/100 = \mathbf{48 \text{ km/h}} \checkmark$$

$$(b) \text{ Net displacement} = 0 \text{ (returns to A)}$$

$$\text{Average velocity} = 0/5 = \mathbf{0 \text{ km/h}}$$

2.3.2 Instantaneous Speed and Instantaneous Velocity

Definitions

Instantaneous Velocity: The velocity at a specific instant of time. It is the limit of average velocity as $\Delta t \rightarrow 0$.

Instantaneous Speed: The magnitude of instantaneous velocity at a given instant. It is what a speedometer reads.

INSTANTANEOUS VELOCITY

$$v = \lim_{\Delta t \rightarrow 0} (\Delta x / \Delta t) = dx/dt$$

The Birth of Calculus — Invented to Describe Motion

The concept of instantaneous velocity required mathematics that didn't exist before the 17th century. How do you find the velocity at a single *instant* when velocity is distance/time — and an instant has zero time?

Isaac Newton (England, ~1665) and **Gottfried Wilhelm Leibniz** (Germany, ~1684) independently invented **calculus** to solve exactly this problem. Newton called his method "fluxions" and used dot notation ($\dot{x}$), while Leibniz developed the dx/dt notation we use today. Their bitter priority dispute lasted decades and divided European mathematics, but both deserve equal credit.

It is remarkable that one of the most powerful branches of mathematics — calculus — was invented largely because physicists needed to describe motion precisely. The derivative dx/dt captures the idea of an instantaneous rate of change, making the concept of "velocity at an instant" mathematically rigorous.

On a **position-time graph**, the instantaneous velocity at any point equals the **slope of the tangent** to the curve at that point. The slope of a secant (chord) between two points gives the average velocity over that interval.

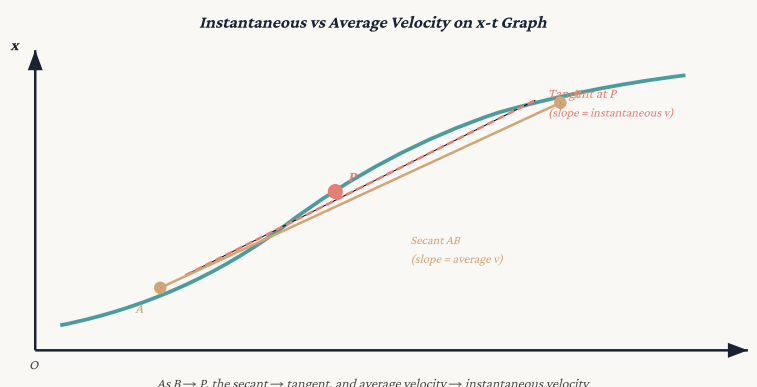


Figure 2.3: Instantaneous velocity (tangent slope) vs average velocity (secant slope) on x-t graph

Solved Example 2.3

Q: The position of an object is given by $x = 3t^2 + 5t + 2$ (m). Find the instantaneous velocity at $t = 2$ s.

Solution:

$$v = dx/dt = d(3t^2 + 5t + 2)/dt = 6t + 5$$

$$\text{At } t = 2 \text{ s: } v = 6(2) + 5 = 12 + 5 = 17 \text{ m/s}$$

2.4 Acceleration

Definition

Acceleration: The rate of change of velocity with respect to time. It is a vector quantity. When acceleration is in the same direction as velocity, the object speeds up; when in the opposite direction, it slows down (deceleration or retardation).

ACCELERATION FORMULAS

$$\text{Average Acceleration: } \bar{a} = \Delta v / \Delta t = (v_2 - v_1) / (t_2 - t_1)$$

$$\text{Instantaneous Acceleration: } a = dv/dt = d^2x/dt^2$$

SI unit of acceleration: m s^{-2} (metre per second squared).

2.4.1 Uniform and Non-Uniform Acceleration

Definitions

Uniform Acceleration: When the velocity of an object changes by *equal amounts in equal intervals of time*, the acceleration is constant (uniform). Example: free fall near Earth's surface ($a = g \approx 9.8 \text{ m/s}^2$).

Non-Uniform Acceleration: When the velocity changes by *unequal amounts in equal intervals of time*, the acceleration varies with time. Example: a car speeding up in city traffic.

Property	Uniform Acceleration	Non-Uniform Acceleration
Acceleration value	Constant (same at all times)	Changes with time
v-t graph	Straight line	Curved line
a-t graph	Horizontal line (constant)	Varying line
Equations of motion	$v = u + at$, $s = ut + \frac{1}{2}at^2$ apply	These equations do NOT apply
Example	Free fall, uniformly braking car	Car in traffic, rocket launch

2.4.2 Instantaneous Acceleration

Instantaneous acceleration is the acceleration at a specific instant. On a **velocity-time graph**, it equals the **slope of the tangent** at that point.

Solved Example 2.4

Q: The velocity of a particle is given by $v = 4t^2 - 2t + 3$ (m/s). Find the acceleration at $t = 3$ s.

Solution:

$$a = dv/dt = d(4t^2 - 2t + 3)/dt = 8t - 2$$

$$\text{At } t = 3 \text{ s: } a = 8(3) - 2 = 24 - 2 = \mathbf{22 \text{ m/s}^2}$$

Velocity Direction	Speed Change	Acceleration Sign	Common Term
Positive (+)	Increasing	Positive (+)	Accelerating
Positive (+)	Decreasing	Negative (-)	Retarding / Decelerating
Negative (-)	Increasing (speed)	Negative (-)	Accelerating (in - direction)
Negative (-)	Decreasing (speed)	Positive (+)	Retarding

Common Misconception

Negative acceleration $\neq$ slowing down always! If an object moves in the negative direction with negative acceleration, it actually *speeds up*. The object slows down only when velocity and acceleration have *opposite* signs.

2.5 Graphical Representation of Motion

Graphs provide a powerful visual tool for analyzing motion. Three types of graphs are particularly useful: position-time (x-t), velocity-time (v-t), and acceleration-time (a-t) graphs. The key insight is that the **slope** and **area** under these graphs carry deep physical meaning.

2.5.1 Position-Time (x-t) Graphs

Properties of x-t Graphs

- **Slope at any point = Instantaneous velocity** at that instant
- Steeper slope = higher speed
- Positive slope = motion in positive direction
- Negative slope = motion in negative direction
- Zero slope (horizontal line) = object at rest
- Constant slope (straight line) = uniform velocity
- Changing slope (curve) = non-uniform velocity (acceleration)

Position-Time Graphs for Different Motions

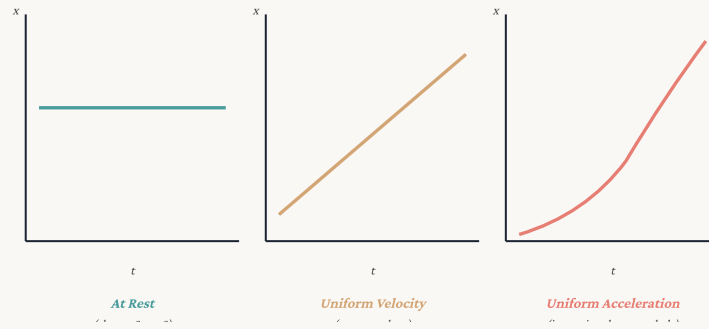


Figure 2.4: Position-time graphs for (a) rest, (b) uniform velocity, (c) uniform acceleration

2.5.2 Velocity-Time (v-t) Graphs

Properties of v-t Graphs

- Slope at any point = Instantaneous acceleration
- Area under the graph = Displacement (signed area — above axis positive, below negative)
- Horizontal line = constant velocity ($a = 0$)
- Straight line with positive slope = constant positive acceleration
- Straight line with negative slope = constant negative acceleration (retardation)
- Line crossing zero = velocity changes direction

Velocity-Time Graphs for Different Motions

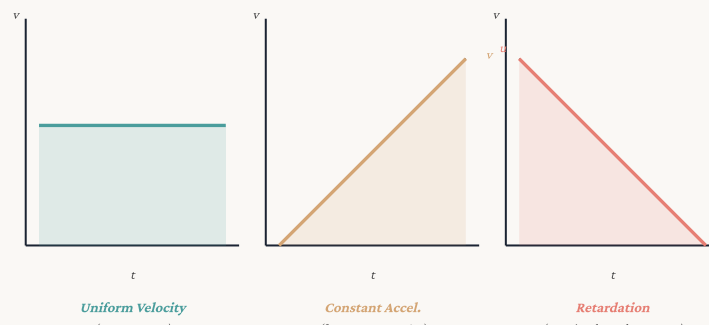


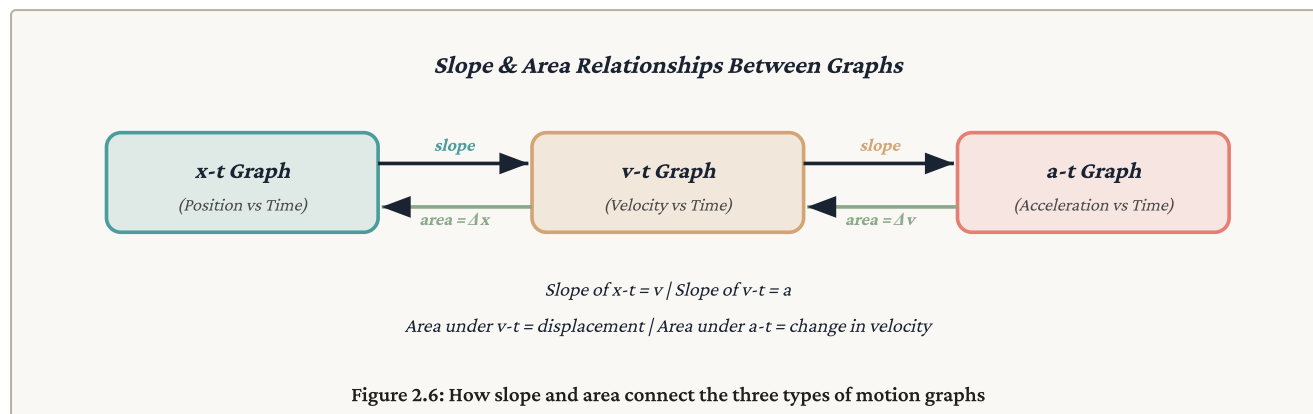
Figure 2.5: Velocity-time graphs — the shaded area equals displacement

2.5.3 Acceleration-Time (a-t) Graphs

Properties of a-t Graphs

- Area under a-t graph = Change in velocity (Δv)
- Horizontal line at $a = \text{constant}$ → uniformly accelerated motion
- Horizontal line at $a = 0$ → uniform velocity
- Horizontal line at $a = -g$ → free fall (upward positive convention)

2.5.4 The Slope and Area Concept — Summary



2.6 Equations of Motion (Uniformly Accelerated Motion)

When an object moves with **constant (uniform) acceleration**, three fundamental kinematic equations relate displacement, velocity, acceleration, and time. These are the most important equations in this chapter.

Variables Used

- **u** = initial velocity (at $t = 0$)
- **v** = final velocity (at time t)
- **a** = constant acceleration
- **t** = time interval
- **s** = displacement during time t

FIRST EQUATION OF MOTION (VELOCITY-TIME RELATION)

$$v = u + at$$

SECOND EQUATION OF MOTION (DISPLACEMENT-TIME RELATION)

$$s = ut + \frac{1}{2}at^2$$

THIRD EQUATION OF MOTION (VELOCITY-DISPLACEMENT RELATION)

$$v^2 = u^2 + 2as$$

DISPLACEMENT IN NTH SECOND

$$s_n = u + a(n - \frac{1}{2}) = u + a(2n - 1)/2$$

2.6.1 Graphical Derivation of Equations

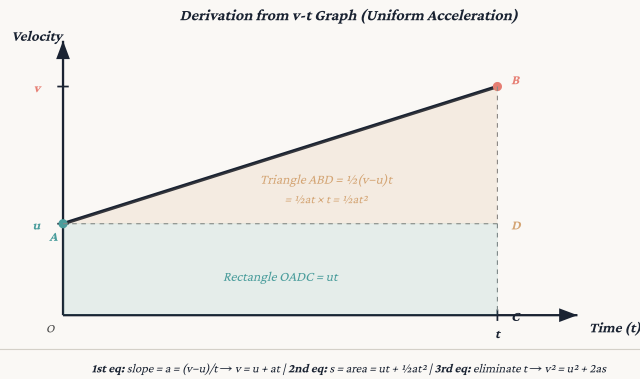


Figure 2.7: Graphical derivation of equations of motion from the v-t graph

💡 Derivation Details

1st Equation: From the v-t graph, slope = acceleration

$$a = (v - u) / t \rightarrow \mathbf{v = u + at}$$

2nd Equation: Displacement = Area under v-t graph = Area of trapezoid OABC

$s = \text{Area of rectangle OADC} + \text{Area of triangle ABD}$

$$s = u \times t + \frac{1}{2} \times (v - u) \times t$$

Since $v - u = at$: $s = ut + \frac{1}{2}(at)(t) \rightarrow \mathbf{s = ut + \frac{1}{2}at^2}$

3rd Equation: From 1st equation: $t = (v - u)/a$. Substituting in 2nd:

$$s = u(v - u)/a + \frac{1}{2}a[(v - u)/a]^2$$

$$s = (uv - u^2)/a + (v - u)^2/2a = (2uv - 2u^2 + v^2 - 2uv + u^2)/2a$$

$$2as = v^2 - u^2 \rightarrow \mathbf{v^2 = u^2 + 2as}$$

When to Use Which Equation

- $\mathbf{v = u + at}$ — When s (displacement) is not involved
- $\mathbf{s = ut + \frac{1}{2}at^2}$ — When v (final velocity) is not involved
- $\mathbf{v^2 = u^2 + 2as}$ — When t (time) is not involved
- $\mathbf{s_n = u + a(n - \frac{1}{2})}$ — For displacement in a specific n th second

💡 Solved Example 2.5

Q: A car accelerates uniformly from rest to 25 m/s in 10 s. Find (a) acceleration, (b) distance covered, (c) distance in the 10th second.

Given: $u = 0$, $v = 25$ m/s, $t = 10$ s

(a) $v = u + at \rightarrow 25 = 0 + a(10) \rightarrow \mathbf{a = 2.5 \text{ m/s}^2}$

(b) $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(2.5)(100) = \mathbf{125 \text{ m}}$

(c) $s_{10} = u + a(n - \frac{1}{2}) = 0 + 2.5(10 - 0.5) = 2.5 \times 9.5 = \mathbf{23.75 \text{ m}}$

💡 Solved Example 2.6

Q: A car accelerates uniformly from 18 km/h to 36 km/h in 5 s. Find (a) acceleration, (b) distance covered.

Solution: First convert: $u = 18 \times 5/18 = 5$ m/s, $v = 36 \times 5/18 = 10$ m/s

(a) $a = (v - u)/t = (10 - 5)/5 = \mathbf{1 \text{ m/s}^2}$

(b) $s = ut + \frac{1}{2}at^2 = 5(5) + \frac{1}{2}(1)(25) = 25 + 12.5 = \mathbf{37.5 \text{ m}}$

Or using $v^2 = u^2 + 2as$: $100 = 25 + 2(1)s \rightarrow s = 75/2 = 37.5 \text{ m} \checkmark$

Solved Example 2.7

Q: A car moving at 50 km/h can stop in 6 m. Find the stopping distance at 100 km/h (same brakes).

Solution: When braking, $v = 0$. Using $v^2 = u^2 + 2as \rightarrow 0 = u^2 + 2as \rightarrow s = u^2/(-2a) = u^2/2|a|$

Since braking force (hence $|a|$) is the same: $s \propto u^2$

$$s_2/s_1 = u_2^2/u_1^2 = (100)^2/(50)^2 = 4$$

$$s_2 = 4 \times 6 = \mathbf{24\text{ m}}$$

Key insight: Doubling the speed quadruples the stopping distance!

2.7 Free Fall — Motion Under Gravity

Definition

Free Fall: Motion of an object under the influence of gravitational force alone, with air resistance neglected. All freely falling objects near Earth's surface experience the same acceleration, called **acceleration due to gravity (g)**, regardless of their mass, shape, or composition.

Galileo and the Leaning Tower of Pisa

Before Galileo: Aristotle taught that heavier objects fall faster — a 10 kg stone should fall 10× faster than a 1 kg stone. This was believed for nearly 2000 years without being tested!

Galileo's Revolution (c. 1590): According to his student Viviani, Galileo dropped two balls of different masses from the Leaning Tower of Pisa and showed they hit the ground at nearly the same time. While this specific story may be legend, Galileo certainly demonstrated the result through careful inclined plane experiments. He showed that **all objects fall with the same acceleration regardless of their mass** (in the absence of air resistance). This was a stunning blow to 2000 years of Aristotelian physics.

Modern confirmation: In 1971, Apollo 15 astronaut **David Scott** dropped a hammer and a feather simultaneously on the Moon (where there is no air). They hit the ground at *exactly* the same time   vindicating Galileo 380 years later, in front of a live television audience!

ACCELERATION DUE TO GRAVITY

$$g \approx 9.8 \text{ m/s}^2 \approx 9.81 \text{ m/s}^2 \text{ (often taken as } 10 \text{ m/s}^2 \text{ for easier calculation)}$$

2.7.1 Sign Convention for Vertical Motion

Convention: Upward = Positive

- Upward displacement: $+s$
- Downward displacement: $-s$
- Upward velocity: $+v$
- Downward velocity: $-v$
- Acceleration due to gravity: $a = -g = -9.8 \text{ m/s}^2$ (always downward)

The equations of motion for free fall become:

EQUATIONS FOR FREE FALL (UPWARD = POSITIVE)

$$v = u - gt$$

$$s = ut - \frac{1}{2}gt^2$$

$$v^2 = u^2 - 2gs$$

2.7.2 Cases of Free Fall

Case 1: Object Dropped from Height ($u = 0$)

Taking downward as positive (for convenience in this case): $a = +g$

$$\text{OBJECT DROPPED FROM REST}$$

$$v = gt \quad | \quad h = \frac{1}{2}gt^2 \quad | \quad v^2 = 2gh$$

Case 2: Object Thrown Vertically Upward

Taking upward as positive: $u = +u_0$, $a = -g$

OBJECT THROWN UPWARD WITH SPEED U

$$\text{Maximum height: } H = u^2/(2g)$$

$$\text{Time to reach max height: } t_{up} = u/g$$

$$\text{Total time of flight: } T = 2u/g$$

Speed on return to start = u (same magnitude, opposite direction)

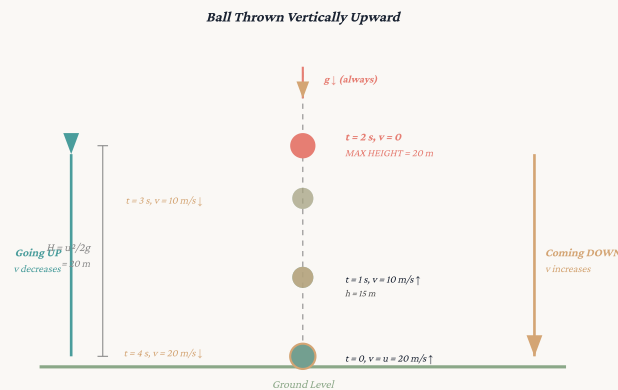


Figure 2.8: Ball thrown upward ($u = 20 \text{ m/s}$, $g = 10 \text{ m/s}^2$): positions, velocities, and times

Solved Example 2.8

Q: A ball is thrown vertically upward with velocity 25 m/s . Find (a) max height, (b) time to reach max height, (c) total time in air. ($g = 10 \text{ m/s}^2$)

(a) At max height, $v = 0$. Using $v^2 = u^2 - 2gH$:

$$0 = 625 - 2(10)H \rightarrow H = 625/20 = \mathbf{31.25 \text{ m}}$$

(b) Using $v = u - gt$: $0 = 25 - 10t \rightarrow t = \mathbf{2.5 \text{ s}}$

(c) Total time = $2t = 2 \times 2.5 = \mathbf{5 \text{ s}}$

Solved Example 2.9

Q: A stone is dropped from the top of a building 80 m high. Find (a) time to reach ground, (b) velocity on hitting. ($g = 10 \text{ m/s}^2$)

Given: $u = 0$, $h = 80 \text{ m}$ (taking downward +)

(a) $h = \frac{1}{2}gt^2 \rightarrow 80 = \frac{1}{2}(10)t^2 \rightarrow t^2 = 16 \rightarrow t = \mathbf{4 \text{ s}}$

(b) $v = gt = 10 \times 4 = \mathbf{40 \text{ m/s}}$

Check: $v^2 = 2gh = 2(10)(80) = 1600 \rightarrow v = 40 \text{ m/s} \checkmark$

⚠ Important Points About Free Fall

- At the **highest point**, velocity = 0 but acceleration = g (NOT zero!). The velocity is momentarily zero, but gravity continues to act
- Time of ascent = Time of descent (if launched and caught at same height)
- Speed at launch = Speed on return to same height
- The equations only apply when air resistance is negligible

2.8 Relative Motion (Basic Concepts)

📘 Definition

Relative Velocity: The velocity of one object as observed from another moving object. If object A has velocity v_A and object B has velocity v_B (both measured from the ground), then the velocity of A *relative to* B is:

RELATIVE VELOCITY IN 1D

$$v_{AB} = v_A - v_B \text{ (velocity of A as seen by B)}$$

$$v_{BA} = v_B - v_A \text{ (velocity of B as seen by A)}$$

Note: $v_{AB} = -v_{BA}$

Cases of Relative Motion

- **Same direction:** $v_{AB} = v_A - v_B$ (relative velocity is the *difference*). Objects appear to move slowly relative to each other
- **Opposite directions:** $v_{AB} = v_A + v_B$ (in magnitude). Objects approach or recede rapidly
- **Both at rest relative to each other:** $v_{AB} = 0$ (when $v_A = v_B$)

Relative Motion — Two Cars



Rain falling vertically appears slanted when you run — that's relative velocity!

Figure 2.9: Relative velocity for objects moving in the same and opposite directions

💡 Solved Example 2.10

Q: Train A moves east at 80 km/h. Train B moves west at 60 km/h on a parallel track. What is the velocity of A relative to B?

Solution: Taking east as positive:

$v_A = +80 \text{ km/h}$, $v_B = -60 \text{ km/h}$ (west = negative)

$v_{AB} = v_A - v_B = 80 - (-60) = 80 + 60 = +140 \text{ km/h}$ (eastward)

A passenger in B sees train A approaching at 140 km/h!

Solved Example 2.11

Q: An athlete runs one complete round of a circular track of diameter 200 m in 40 s. Find (a) distance, (b) displacement, (c) average speed, (d) average velocity.

Solution:

- (a) Distance = circumference = $\pi d = \pi \times 200 = 200\pi \approx 628.3 \text{ m}$
- (b) Displacement = 0 (returns to starting point) $\rightarrow 0 \text{ m}$
- (c) Average speed = distance/time = $628.3/40 = 15.7 \text{ m/s}$
- (d) Average velocity = displacement/time = $0/40 = 0 \text{ m/s}$

Chapter Summary — Must Remember

- Rest and motion are **relative** — they depend on the frame of reference
- Distance is scalar (≥ 0); displacement is vector (can be +, −, or 0)
- $|\text{Displacement}| \leq \text{Distance}$ always; equal only for straight-line one-way motion
- Average speed = total distance / total time; Average velocity = displacement / time
- Instantaneous velocity = dx/dt = slope of x-t graph at that point
- Acceleration = dv/dt = slope of v-t graph; can be uniform or non-uniform
- Equations of motion ($v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$) apply only for **constant acceleration**
- Slope of x-t = velocity; Slope of v-t = acceleration
- Area under v-t = displacement; Area under a-t = change in velocity
- Free fall: $g \approx 9.8 \text{ m/s}^2$ downward; at max height $v = 0$ but $a = g$
- Time up = time down (same launch/catch height); speed at return = launch speed
- Relative velocity: $v_{AB} = v_A - v_B$

Common Mistakes to Avoid

- Using distance when displacement is required (and vice versa)
- Forgetting sign conventions — be consistent!
- Assuming negative acceleration always means slowing down
- Using equations of motion for non-uniform acceleration
- Confusing average speed with average velocity
- Forgetting that at maximum height, $v = 0$ but $a = g \neq 0$
- Not converting units (km/h to m/s: multiply by 5/18)
- Mixing up u and v in equations

Practice Questions (Unsolved)

Conceptual Questions

1. Can an object have zero velocity and non-zero acceleration? Example?
2. Does negative acceleration always mean slowing down? Explain.
3. Why is velocity zero at highest point but acceleration is not?
4. When is average speed equal to magnitude of average velocity?
5. Explain relative velocity with a real-life example.
6. What does the area under a v-t graph represent?
7. Distinguish between distance and displacement with a round-trip example.

Short Answer Questions (2-3 Marks)

1. Derive $v = u + at$ from the velocity-time graph.
2. Show that displacement in n-th second: $s_n = u + a(n - \frac{1}{2})$.
3. Define instantaneous velocity. How is it different from average velocity?
4. What information can you get from a v-t graph? List at least 4 things.

Numerical Problems

1. Car accelerates from 18 km/h to 36 km/h in 5 s. Find (a) acceleration, (b) distance.
2. Stone dropped from 80 m. Find time and hitting velocity. ($g = 10 \text{ m/s}^2$)
3. Ball thrown up at 25 m/s. Find max height, time to top, total time. ($g = 10 \text{ m/s}^2$)
4. Athlete completes circular track ($d=200 \text{ m}$) in 40 s. Find distance, displacement, avg speed, avg velocity.
5. Car at 50 km/h stops in 6 m. Stopping distance at 100 km/h?
6. Train from rest reaches 72 km/h in 5 min. Find acceleration and distance.
7. Two trains approach at 60 km/h and 40 km/h on parallel tracks. Relative velocity?

Assertion-Reason Questions

(a) Both true, R explains A (b) Both true, R not explanation (c) A true, R false (d) A false, R true

1. **A:** Displacement can be zero when distance is non-zero. **R:** Displacement depends only on initial and final positions.
2. **A:** Equations of motion apply to any motion. **R:** They are derived only for uniform acceleration.
3. **A:** At highest point of throw, acceleration is zero. **R:** Velocity is zero there.
4. **A:** Average speed $\geq$ |average velocity| always. **R:** Distance $\geq$ |displacement| always.